

TOTAL

26.5 / 30

COMMUNICATION: (4 Marks)

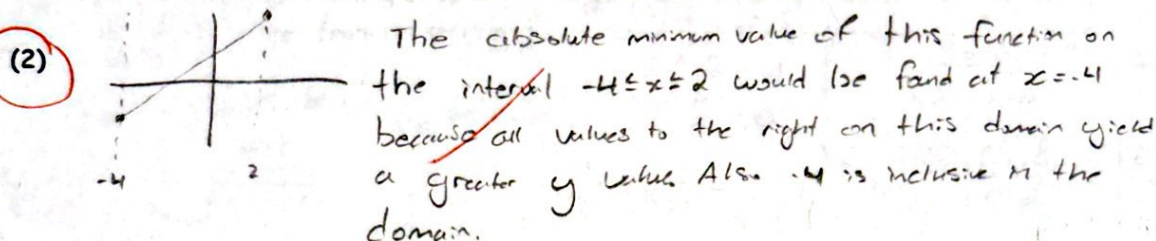
1. The position of an object moving along a straight line can be modeled by the function $s(t) = 10 + 6t - t^2$ where s is the position in metres and t is the time in seconds, $t \geq 0$. Is the object speeding up or slowing down at 5 seconds? Justify your response.

(2) $s(t) = 10 + 6t - t^2$
 $t \geq 0$
 $s'(t) = v(t) = 6 - 2t$
 $v'(t) = a(t) = -2$

$v(5) = -4$
 $a(5) = -2$

Since $v(5) < 0$ and $a(5) < 0$, the object is moving in a negative direction and also accelerating in a negative direction, so the object is speeding up at 5 seconds.

2. A function increases on the interval $-4 \leq x \leq 2$. Where would you specifically find the absolute minimum value? Include a diagram as part of your explanation.



KNOWLEDGE: (11 Marks)

1. Determine the exact absolute maximum and minimum points for $f(x) = \frac{4x}{x^2 + 1}$, $-2 \leq x \leq 4$.

(3)

$f(x) = \frac{4x}{x^2 + 1}$
 $-2 \leq x \leq 4$
 $f(-2) = -\frac{8}{5}$
 $f(-1) = -2$
 $f(1) = 2$
 $f(4) = \frac{16}{17}$

$f'(x) = \frac{4(x^2 + 1) - 4x(2x)}{(x^2 + 1)^2}$
 $= \frac{4x^2 + 4 - 8x^2}{(x^2 + 1)^2}$
 $= \frac{-4x^2 + 4}{(x^2 + 1)^2} \rightarrow \text{C.P. } x = \pm 1$

Max: (1, 2)
 Min: (-1, -2)

2. A particle moves along a line. The particle's distance from a fixed point, s , in metres as a function of time, t , in seconds can be modelled by $s(t) = \frac{2t^3}{3} - \frac{13t^2}{2} + 6t + 1$, $t \geq 0$.

- a) When does the particle change direction? Include an appropriate sign chart.

(2)

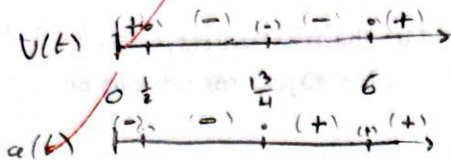
$s(t) = \frac{2t^3}{3} - \frac{13t^2}{2} + 6t + 1$
 $s'(t) = v(t) = 2t^2 - 13t + 6$
 $0 = 2t^2 - 13t + 6$
 $0 = (2t - 1)(t - 6)$
 $2t - 1 = 0 \rightarrow t = 0.5$
 $t - 6 = 0 \rightarrow t = 6$

C.P. $x = 0.5, 6$
 $t \geq 0$
 particle changes direction when $v(t) = 0$, and it has to change signs. eg. can't be: $\uparrow \downarrow$
 $\therefore$ Particle changes direction at 0.5 seconds and 6 seconds.

b) For what interval(s) is the particle slowing down? Include any appropriate sign charts.

(2) $V(t) = 2t^2 - 13t + 6 = (2t-1)(t-6)$

$V'(t) = a(t) = 4t - 13 \rightarrow t = \frac{13}{4}$
 $t \geq 0$



Particle slows down when velocity and acceleration are different signs.

Therefore particle is slowing down on intervals:

$[0, \frac{1}{2}) \cup (\frac{13}{4}, 6)$

3. Determine the exact equation of the normal to given $x^2 - 3xy + y^2 = -1$ at (1, 2).

(4) $x^2 - 3xy + y^2 = -1$
 $2x - (3y + 3x(\frac{dy}{dx})) + 2y(\frac{dy}{dx}) = 0$

$-3x(\frac{dy}{dx}) + 2y(\frac{dy}{dx}) = 3y - 2x$

$\frac{dy}{dx}(2y - 3x) = 3y - 2x$

$\frac{dy}{dx} = \frac{3y - 2x}{2y - 3x}$

$\left. \frac{dy}{dx} \right|_{\substack{x=1 \\ y=2}} = \frac{3(2) - 2(1)}{2(2) - 3(1)}$

$m = \frac{6-2}{4-3}$

$m = 4 \rightarrow \text{tangent}$

$m_{\text{normal}} = -\frac{1}{4}$

$y = -\frac{1}{4}x + b$

$2 = -\frac{1}{4}(1) + b$

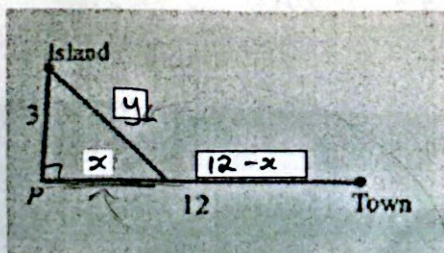
$b = 2 + \frac{1}{4}$

$b = \frac{9}{4}$

$y = -\frac{1}{4}x + \frac{9}{4}$

APPLICATION: (10 Marks)

1. A small island is 3km from the nearest point P on the straight shoreline of a large lake. A town is 12km down the shore from P. See the figure. If a person on the island can row a boat 2.5km/h and can walk 4km/h, where should the boat be landed so that the person arrives in town in the shortest time?



let $T = \text{time}$
 x be this length
 y be this

$T = 4\left(\frac{12-x}{4}\right) + \frac{5}{2}\left(\frac{y}{2.5}\right)$

$y^2 = x^2 + 3^2$

(5) $y = \sqrt{x^2 + 9}$

$T = 4(12-x) + \frac{5}{2}\sqrt{x^2 + 9}$

$T = 48 - 4x + \frac{5}{2}\sqrt{x^2 + 9}$

$T' = -4 + \frac{5}{4}(x^2 + 9)^{-\frac{1}{2}}(2x)$

domain $0 \leq x \leq 12$

$T' = -4 + \frac{10x}{4\sqrt{x^2 + 9}}$

$4 = \frac{10x}{4\sqrt{x^2 + 9}}$

$4\sqrt{x^2 + 9} = \frac{10x}{4}$

$\sqrt{x^2 + 9} = \frac{10x}{16}$

$x^2 + 9 = \frac{100x^2}{256}$

$x^2 + 9 = \frac{25}{64}x^2$

$9 = -\frac{39}{64}x^2$

$14.77 = x^2$

$x = 3.84$

The boat should be landed 3.84 km from point P.

check ends!

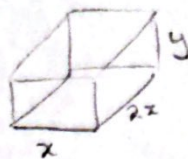
9.5

2. A woman wants to construct a box whose base length is twice the base width. The material to build the top and bottom is \$9/m² and the material to build the sides is \$6/m². If the woman wants the box to have a volume of 70 m³, determine the dimensions of the box (in metres) that will minimize the cost of production. What is the minimum cost?

Don't

(5)

4.5



$$V = 70 \text{ m}^3$$

$$V = 2x^2y = 70$$

$$y = \frac{35}{x^2}$$

Let $C(x)$ cost

$$C(x) = 2(9)(2x^2) + 2(6)(2xy) + 2(6)xy$$

$$C(x) = 36x^2 + 24xy + 12xy$$

$$C(x) = 36x^2 + \frac{840}{x} + \frac{420}{x}$$

$$= 36x^2 + \frac{1260}{x}$$

$$C'(x) = 72x + (-\frac{1260}{x^2})$$

$$= x^{-2}(72x^3 - 1260)$$

$$= \frac{12(6x^3 - 105)}{x^2}$$

$$C': x = \sqrt[3]{\frac{105}{6}}$$

$$x = 2.60$$

$$\text{Width: } 2.60 \text{ m}$$

$$\text{Length: } 5.20 \text{ m}$$

$$\text{height: } 5.18 \text{ m}$$

$$\text{Cost: } \$728.21$$

THINKING: (5 Marks)

1. The velocity of a falling body is $v = 4\sqrt{25-6t} + 3$ feet per second. At the instant t seconds the body has fallen s feet from its starting point. Show that the acceleration is 16 ft/sec^2

$$a = 2(25-6t)^{-\frac{1}{2}}$$

$$= \frac{-12t}{\sqrt{25-6t}}$$

$$16\sqrt{25-6t} = -12t$$

$$(2) a(s) = \frac{-12t}{\sqrt{25-6t}}$$

$$16 = \frac{-12t}{\sqrt{25-6t}}$$

2. Let $y = \frac{2}{3}(x-a)^2 - \frac{1}{4}(x-b)^2$. Determine the x -value(s) for which there could be a local extrema.

$$y = \frac{2}{3}(x-a)^2 - \frac{1}{4}(x-b)^2$$

$$(3) y' = \frac{4}{3}(x-a) - \frac{1}{2}(x-b)$$

$$0 = \frac{4}{3}x - \frac{4}{3}a - \frac{1}{2}x + \frac{1}{2}b$$

$$\left(\frac{4}{3}a - \frac{1}{2}b\right) = \left(\frac{4}{3}x - \frac{1}{2}x\right)$$

$$8a - 3b = 8x - 3x$$

$$8a - 3b = 5x$$

$$x = \frac{8a-3b}{5}$$

8